Silas Bambil

Atlanta, Georgia | 404-468-7274 | sdbambil98@gmail.com

Education

Georgia State University August 2026 to Present
MS Mathematics, concentration in Statistics | Atlanta, Georgia
Expected graduation: spring 2028

Georgia State University May 2025
MS Quantitative Risk Analysis and Management | Atlanta, Georgia

Kwame Nkrumah University of Science and Technology July 2020
BSc Actuarial Science | Kumasi, Ghana

Research Interests

Extreme value theory and tail-risk modeling; catastrophe risk and insurance; statistical prediction and model evaluation; distribution shift and model adaptation; credit risk; Monte Carlo methods.

Research Experience

Beyond Quantile Agreement: Decomposing Expected Shortfall Discrepancies in Fire-Loss Models 2026
Research manuscript in preparation with Shudeshna Nandy, Clement Ampong, and Prakash Karki. Code and reproducibility archive: Zenodo, DOI 10.5281/zenodo.23253824.

- Formulated the research question and designed the methodology to examine discrepancies between empirical and modeled Expected Shortfall in severe fire losses.
- Analyzed approximately 1.25 million positive-loss U.S. structure-fire records from the National Fire Incident Reporting System, 2016–2025.
- Applied peaks-over-threshold methods and generalized Pareto models; assessed sensitivity to thresholds, alternative tail models, and event-level aggregation.
- Examined the contribution of exceptionally large losses to Expected Shortfall discrepancies and used bounded-layer analyses to assess uncertainty in the far tail.

Predictive Stability and Adaptation Under Distribution Shift 2026
Research manuscript, unpublished.

- Studied calibration, discrimination, probability accuracy, and decision thresholds using a loan-default benchmark and controlled Monte Carlo experiments.
- Conducted 500 replications per condition in R to examine covariate shift with model misspecification, baseline-risk changes, and changes in predictor effects.
- Compared recalibration, refitting, and threshold revision using independent evaluation samples.

Greenberg Sprint Predictive Enrollment Model 2024
Graduate research project | Georgia State University

- Contributed to a predictive model of graduate enrollment using behavioral and demographic data.
- Cleaned applicant data, developed predictor variables, and supported model evaluation for graduate recruitment analysis.

Application of Extreme Value Theory to Fire Insurance 2020
Undergraduate thesis | Kwame Nkrumah University of Science and Technology

- Examined extreme value methods for modeling severe fire-insurance losses and tail-risk behavior.

Teaching Experience

Georgia State University August 2026 to Present
Graduate Lab Assistant | Atlanta, Georgia

- Support students in College Algebra (MATH 1111) and Support for College Algebra (MATH 0999), emphasizing mathematical reasoning and problem solving.

Osceola School for the Arts January 2026 to June 2026
AP Mathematics Teacher | Osceola County, Florida

- Taught AP Calculus, AP Precalculus, and AP Statistics; prepared lessons and assessments.
- Achieved a 70% AP examination pass rate among students taught.

Professional Experience

Georgia State University August 2023 to June 2025
Graduate Assistant | Atlanta, Georgia

Supported predictive enrollment analysis and faculty risk-analysis projects through statistical analysis, research assistance, and data management.

OPH CHEM CONSULT May 2022 to August 2023
Finance Associate | Ghana

Conducted financial research on misappropriation risks and community financial challenges and contributed to analytical reports and briefs.

Ghana Statistical Service September 2020 to August 2021
Research Assistant | Ghana

Contributed research, statistical analysis, and data-management support to population and economic studies.

Technical Skills

Programming and software: R, Python, SAS, Microsoft Excel, and Microsoft Office Suite.

Statistical methods: logistic regression, multiple regression, Monte Carlo simulation, predictive model evaluation, probability calibration, bootstrap methods, data cleaning, and statistical visualization.

Risk and actuarial methods: extreme value theory, peaks-over-threshold modeling, generalized Pareto models, Value-at-Risk, Expected Shortfall, credit-risk analysis, stress testing, and scenario analysis.

Awards and Certifications

Graduate Merit Scholarship, Department of Mathematics and Statistics, Georgia State University | August 2026

Graduate Merit Scholarship, Robinson College of Business, Georgia State University | August 2023 to May 2025

FINRA Securities Industry Essentials (SIE) Examination | January 2025

FINRA Uniform Securities Agent State Law Examination (Series 63) | August 2025